RETURNGUARD

From Delivery to Return — Trust Every Step- ZENLY

THE PROBLEM

$101B

Annual return fraud losses globally

10.7%

Of all retail returns are fraudulent

68%

Retailers lack robust verification

WARDROBING
Using items then returning them

ITEM NOT RECEIVED
False delivery dispute scams

ORGANIZED FRAUD RINGS
Coordinated large-scale return abuse

FAKE RETURNS
Returning different/damaged products

RECEIPTMANIPULATION
Forged or altered purchase proofs

RECEIPTMANIPULATION
Forged or altered purchase proofs

WHERE THE SYSTEM BREAKS DOWN TODAY

CURRENT GAPS

No Condition Proof at Delivery

No photographic or video evidence of product state when received. Retailers cannot verify original condition.

No Delivery Authentication

OTP and geolocation checks are absent. Packages can be marked delivered without actual recipient confirmation.

Heavy Financial Losses

Retailers absorb 100% of fraud costs. Loss of product + refund + operational overhead with no recourse.

Weak Return Validation

Current processes rely on customer self-reporting. There is no authentic cross-verification against delivery data.

RETURNGUARD

First Unboxing Video

Mandatory video proof of product condition at moment of opening

QR Smart Packaging

Unique tamper-proof QR codes linked to every order for instant verification entry

AI Return Validation

Computer vision and ML analyze images, EXIF data and fraud patterns

OTP + Location Delivery

Geofenced one-time password confirmation for every delivery event

RETURNGUARD TECHNICAL PIPELINE WORKFLOW

LAYER 1: SIGNAL COLLECTION (CLIENT-SIDE) [cite: 1]

- **Conversational UI (`ReturnChat.jsx`)** [cite: 1]
- **Linguistic Signals:** Description vs Reason
- **Behavioral Telemetry:** Dwell Time, Flight Time, Mouse Velocity, Scroll Rhythm
- **Device & Network Fingerprinting:** Fingerprint Hash, IP
- **Live Camera Anti-Spoofing (`DamagedProductCapture.jsx`):** gallery block, `getUserMedia`, Metadata Embedding [cite: 1]

LAYER 2: FEATURE EXTRACTION (BACKEND)

FastAPI Backend (`main.py`): Data Normalization, Delta Calculation [cite: 1]
Historical Context Injection: Supabase `claims` table lookup (prior_return_count, account_id) [cite: 1]

LAYER 3: PARALLEL SCORERS [cite: 1]

- **TELEMETRY SCORER (Bot/Script Detection):** dwell <= 30ms, flight > 250ms, mouse >= 800px/ms, scroll <= 10ms variance [cite: 1]
- **LINGUISTIC & NLP SCORER (Wardrobing/Misrepresentation):** Reason vs Description Contradiction, Vagueness, Minimal Effort [cite: 1]
- **NETWORK & HISTORY SCORER (Organized Rings):** Shared Fingerprints, VPN/Proxy Subnets, Serial Return Patterns, Friendly Fraud/INR Markers [cite: 1]

LAYER 4: THE FUSION ENGINE (CORROBORATION) [cite: 1]

- Cumulative Scoring Accumulation (0-100) [cite: 1]
- **Strict Corroboration Rule:** Score cannot escalate above AMBER without at least 2 independent signals triggering. [cite: 1]

LAYER 5: DECISION TIER & RESILIENCY [cite: 1]

Decision Map

Score Range	Action	System Response [cite: 1]
GREEN (0-30)	Auto-Approve	Instant refund. Invisible to user. [cite: 1]
AMBER (31-60)	Friction Added	Live photo capture required. [cite: 1]
ORANGE (61-80)	Human Review	Sent to admin queue. 24h SLA. [cite: 1]
RED (81-100)	Block & Appeal	Immediate rejection. [cite: 1]

RESILIENT ARCHITECTURE (`llm_service.py`) [cite: 1]

- **Rate Limiting:** 15 req/min in-memory bucket [cite: 1]
- **Caching:** 10-minute TTL for identical payloads [cite: 1]
- **Exponential Backoff:** Retries on transient errors [cite: 1]
- **Deterministic Fallback:** Fallback to pure Python heuristics if LLM unavailable [cite: 1]

IMPACTS

1

Detect and Prevent Fraud

Identifies and stops various fraud types

2

Identify Fraud Rings

Uncovers organized fraud networks

3

Improve Decision Accuracy

Enhances fraud classification decisions

4

Accelerate Investigation

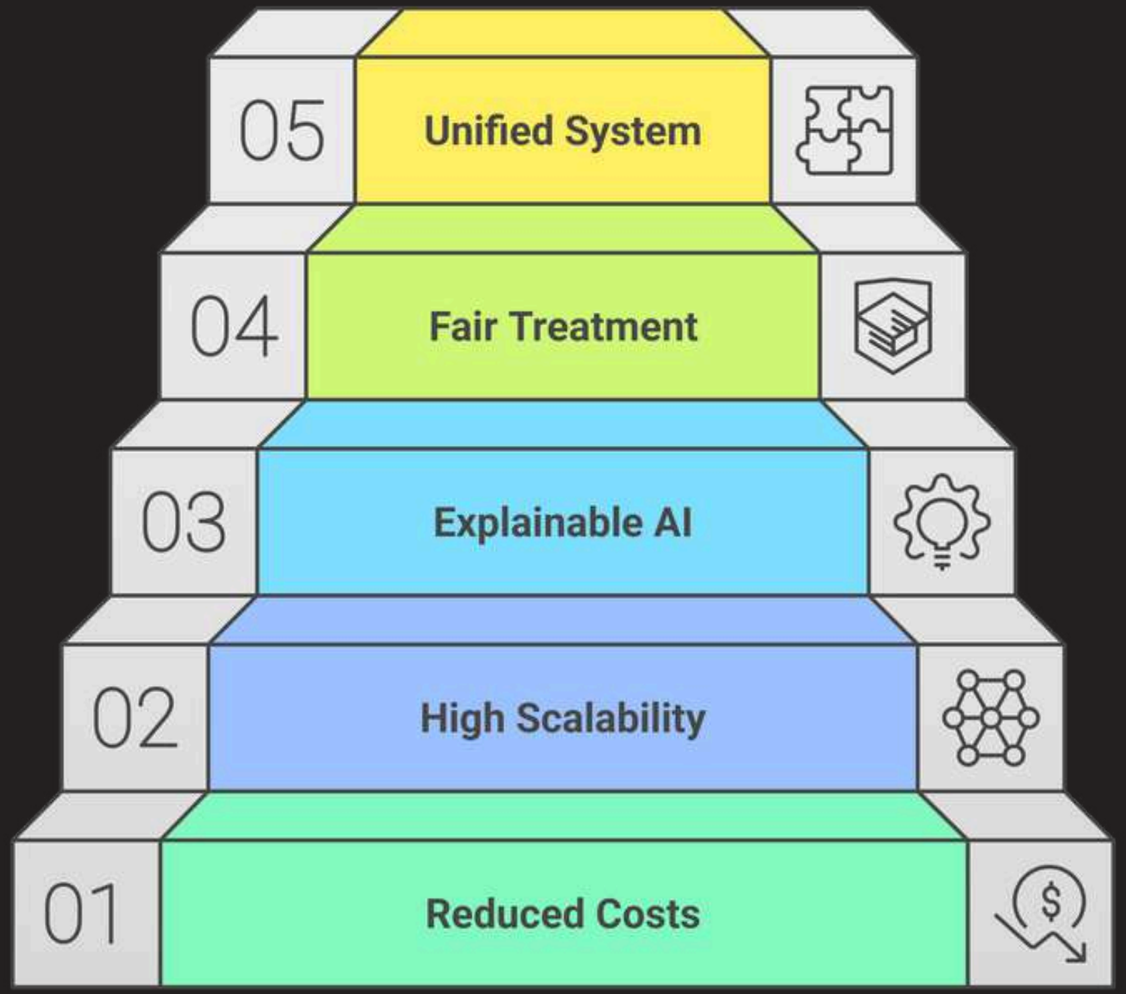
Reduces time for fraud investigation

5

Strengthen Security and Trust

Increases platform integrity and trust

BENIFITS



TARGET CUSTOMER

E-commerce Marketplaces
Platforms like Amazon-like marketplaces with multi-seller ecosystems and high return volume.

Fashion & Apparel Brands
Clothing retailers and D2C fashion startups facing wardrobing and high return rates.

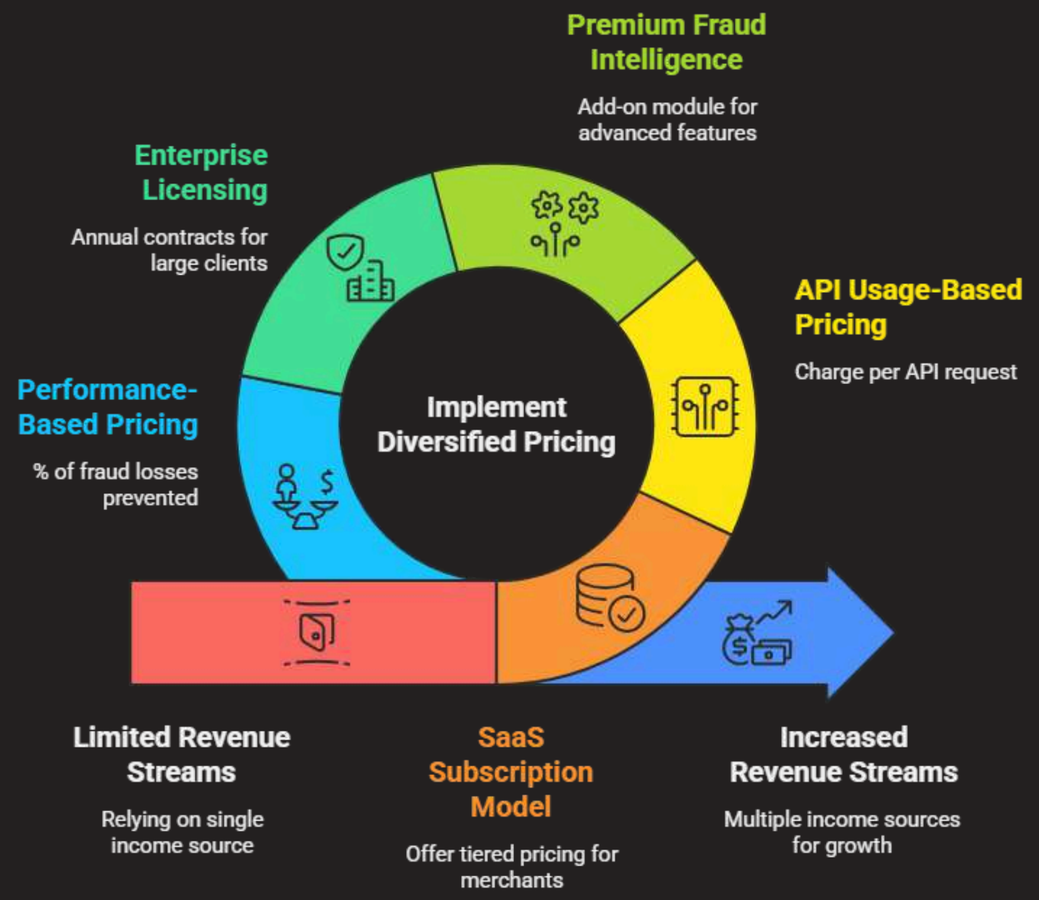
Electronics Retailers
Sellers of smartphones, gadgets, and accessories dealing with high-value products and friendly fraud.

Logistics & Delivery Companies
Last-mile delivery partners experiencing INR disputes and delivery fraud claims.

Payment Gateways / Fintechs
Razorpay-like platforms and BNPL services needing chargeback detection and transaction risk scoring.

Enterprise Retail Chains
Offline + online hybrid stores and omnichannel retailers requiring unified fraud analytics.

BUSSINESS MODEL



RETURNGUARD

"From Delivery to Return — Trust Every Step"

- End-to-end trust from delivery to return
- Protects retailers AND genuine customers
- AI-powered, scalable, and integration-ready
- Measurable ROI from Day 1

Ready to Eliminate Return Fraud?

https://github.com/navvwd/HACK_HUSTLE

THANK YOU